ABSTRACT OF THE DISCLOSURE

A system for a Central Processing Unit (CPU) that is natively designed to process ultra-wide data words received directly from an integrated, massively parallel optical interface. The system comprises an execution core with arithmetic logic units (ALUs), registers, and instruction decoders specifically architected to operate on data words of 256 bits or greater. This execution core is directly coupled to an optical I/O unit on the same die, eliminating the need for traditional serial-to-parallel conversion logic and other bottlenecks associated with electrical I/O. This optically-native design allows the CPU to ingest and process data at the full line rate of an ultra-high-bandwidth optical bus, enabling unprecedented computational throughput.